OtisHealth

OtisHealth Response to Request for Information, Health Technology Ecosystem
Department of Health and Human Services, Centers for Medicare & Medicaid Services
RIN 0938–AV68 | File code CMS–0042–NC

Dear Secretary Kenndey, Dr. Oz, and Dr. Keane.

Thank you for giving us an opportunity to contribute our perspective as application developers, patients, and caregivers. We hope our comments are helpful as you consider policies and initiatives to encourage innovation that enhances the care of Medicare beneficiaries and reduces healthcare costs. We welcome further dialogue on any of these important topics.

Sincerely,
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B. PATIENTS AND CAREGIVERS	
1. Patient Needs	
PC–1. What health management or care navigation apps would help you understand and manage your (or your loved ones) health needs, as well as the actions you should take?	As both a patient and caregiver, I would find a health management app helpful if it guided me in collecting baseline health data, setting reminders for medications and vitals, providing insights based on health inputs, and suggesting topics to discuss with my healthcare provider.
a. What are the top things you would like to be able to do for your or your loved ones' health that can be enabled by digital health products?	**Top priorities for digital health tools:** • A unified app to collect personal health records from all sources • The ability to annotate records (e.g., flag incorrect diagnoses, note discontinued meds) and share with providers • Add and share caregiver observations, vitals, mental health assessments, and daily living activities
b. If you had a personal assistant to support your health needs, what are the top things you would ask them to help with?	1 – Based on my current health data and my family history, do I have any health concerns I should discuss with my doctor? 2 – Am I missing any tests or data that would help me better understand my health? 3 – Is my loved one adhering to their care plan and maintaining their health?

PC–2. Do you have easy access to your own and all your loved ones' health information in one location (for example, in a single patient portal or another software system)?	No.
b. If not, in what contexts or for what workflows would it be most valuable to such health information?	When a loved-one needs to see a new healthcare provider, it would be very beneficial to be able to walk in the door of the new healthcare provider with all my loved-one's health data, from all sources, ready to share. This would ensure that the healthcare provider has all relevant and current health data before proceeding with additional testing, diagnoses, and treatment. When a loved-one is receiving care, it is essential to see any updates to their health information, as it's available.
c. Were there particular data types, such as x-rays or specific test results, that were unavailable? What are the obstacles to accessing your own or your loved ones' complete health information electronically and using it for managing health conditions or finding the best care (for example, limitations in functionality, user friendliness, or access to basic technology infrastructure)?	A major obstacle is that I need login credentials for every provider portal in order to fetch my records or the records of a loved-one. So even if I use an app to collect my data, I still have to log into each portal to retrieve my records. A larger obstacle is that not all provider EHRs send the data in a standardized format. So the information shown in records for clinical visits, procedures, and diagnoses vary from one provider to the next.
PC–3. Are you aware of health management, care navigation, or personal health record apps that would be useful to Medicare beneficiaries and their caregivers?	Yes. OtisHealth (https://otishealth.net) provides mobile and web apps for families and caregivers that allows them to collect health data from different sources, manually enter data, set medication reminders, connect to wearable devices, connect to home health monitoring devices, and securely share with family members. OtisHealth is a TEFCA Individual Access Services (IAS) vendor, partnered with Health Gorilla (https://www.healthgorilla.com/) to provide access through the national exchange framework. This would give patients and potentially their caregivers access to their health records, wherever they are stored, and without additional login credentials. Specific to Medicare beneficiaries, OtisHealth launched an inexpensive Remote Physiologic Monitoring platform (https://otishealth.net/rpm/) that provides a simple, low-overhead service to CMS medical practices and their patients.

PC–4. What features are missing from apps you use or that you are aware of today?	I would like to send my combined health records, including my self-reported health information, directly to my doctor's EHR from my personal health app. This would help me avoid filling out medical history forms at a new or existing doctor's office, where it's likely I will forget essential information.
a. What apps should exist but do not yet? Why do you believe they do not exist yet?	I would like an app that allows me to pick a provider from a list of ALL providers and fetch my records without having to log into every provider portal. I would love it if the app also allowed me as a family caregiver to fetch my Mom's healthcare records without having to set-up individual accounts and sign forms with each provider to access her records. These apps may not yet exist because healthcare providers have not widely adopted digital identity verification methods. Additionally, caregiver access remains limited due to the absence of national standards for associating digital credentials, such as a digital Medical Power of Attorney (dMPOA), between patients and caregivers.
b. What set of workflows do you believe CMS is uniquely positioned to offer?	It would be wonderful if CMS could open its systems to allow a patient or caregiver using their own app to retrieve their CMS health record and claims without having to log into CMS directly.
PC–5. What can CMS and its partners do to encourage patient and caregiver interest in these digital health products?	CMS could raise awareness by launching campaigns that explain programs like Remote Physiologic Monitoring, which help patients with chronic conditions stay healthier and avoid hospital visits. More importantly, CMS should encourage providers to enable direct, low-burden sharing of health data with patients' personal health apps, increasing adoption and engagement.

a. What role, if any, should CMS have in reviewing or approving digital health products on the basis of their efficacy, quality or impact or both on health outcomes (not approving in the sense of a coverage determination)? What criteria should be used if there is a review process? What technology solutions, policy changes, or program design changes can increase patient and caregiver adoption of digital health products (for example, enhancements to data access, reimbursement adjustments, or new beneficiary communications)?	It may be helpful for CMS to simply catalog the available digital health products and their declared offering, capabilities, stated impact, regulatory compliance, accreditations, and certified services. CMS could verify that the company that owns the digital health product is legally incorporated (may use Dunn & Bradstreet or IRS) and their catalog listing includes links to all the claims they have made (e.g. links to their accreditation documentation). From a policy and program perspective, CMS could expand reimbursement for digital tools used in chronic care management, behavioral health and remote patient monitoring. For example for Remote Physiologic Monitoring (RPM), Blood Glucose Meters (BGM) and Continuous Glucose Monitors (CGM) are covered. However, blood pressure cuffs, pulse oximeters, spirometers, and tonometers are not. Based our experience with RPM by OtisHealth, expanding the reimbursement would improve RPM uptake by providers who currently must purchase these devices for their patients and thus improve the potential patient health benefits and the financial benefits to the government. It would also be helpful for CSM to sponsor workshops tailored to older adults to teach them how they can utilize digital health products to improve their health.
PC–6. What features are most important to make digital health products accessible and easy to use for Medicare beneficiaries and caregivers, particularly those with limited prior experience using digital tools and services?	Digital health products should be intuitive, with customizable accessibility features—such as adjustable button sizes, voice assistance, text transcripts, and language options—to support patients of all backgrounds. In certain situations, it would be helpful for a caregiver to be able to help the patient with the use of digital tools. For example, OtisHealth's Alzheimer's management tool allows caregivers to complete and submit assessments on behalf of patients.

PC–7. If CMS were to collect real- world data on digital health products' impact on health outcomes and related costs once they are released into the market, what would be the best means of doing so?	CMS could ask the digital health products companies to submit studies that can be published with their product listing that shows its success in improving outcomes. CMS could evaluate RPM's impact by tracking emergency department visits over a one-year period among patients using RPM tools compared to those not enrolled. Ensuring that monitored programs include compliance tracking and provider engagement would improve study validity.
2. Data Access and Integration	
PC–8. In your experience, what health data is readily available and valuable to patients or their caregivers or both?	Lab results, appointments, medication history, visit summaries, billing statements.
a. What data is valuable, but hard for patients and caregivers, or app developers and other technical vendors, to access for appropriate and valuable use (for example, claims data, clinical data, encounter notes, operative reports, appointment schedules, prices)?	Patients often struggle to have complete pricing transparency (i.e. price of visit or procedure beforehand and what is covered by insurance). Patients also struggle to build a summary of care from different providers on one platform. Clinical notes and encounter summaries are also a challenge to find and are often not shared.
b. What are specific sources, other than claims and clinical data, that would be of highest value, and why?	Patient reported data on outcomes outside of clinical settings would be of incredible value as it would aid in treatment effectiveness (i.e. no one can monitor your health as frequently as yourself).
c. What specific opportunities and challenges exist to improve accessibility, interoperability and integration of clinical data from different sources to enable more meaningful clinical research and generation of actionable evidence?	The best opportunity is for patients and their caregivers to be at the center of the patient's health data. They should be able to aggregate all their data from all sources, be able to add their own observations and data from devices, and then securely share with anyone that needs to see it. As the collection of data is highly fragmented in terms of interoperability compliance, patient access options, and data format, the aggregation of the data is a challenge today.
PC–10. How is the Trusted Exchange Framework and Common Agreement™ (TEFCA™) currently helping to advance patient access to health information in the real world?	TEFCA Individual Access Services (IAS) capability is the most important feature for patients to have unfettered access to their health records. In particular, the use of federated digital credentials (IAL2) to enable queries with portal logins is the most critical aspect of TEFCA IAS that will provide the best access to patients, nationwide.

a. Please provide specific examples.	The best example of the power of IAL2-enabled TEFCA IAS queries is I, as a patient, can use my personal health app (OtisHealth) to search for and retrieve my health records from any healthcare provider I may have seen, anywhere in the USA, without having to remember where I was seen and without having to log into each EHR patient portal.
b. What changes would you suggest?	Require all healthcare providers to join the network through a QHIN (or an HIE that is connected to a QHIN) and require them to accept IAL2-enabled queries.
c. What use cases could have a significant impact if implemented through TEFCA?	A significant use case would be enabling a family caregiver the ability to request a patient's records using the caregiver's digital identity that was associated with the patient's digital identity. In this way, the proper record of the request origin can be maintained, and a caregiver can get access to the records they need to care for their loved-one.
d. What standards are you aware of that are currently working well to advance access and existing exchange purposes?	The adoption of NIST 800-63-3 IAL2/AAL2 has really opened the door for more efficient and secure patient and provider digital credential issuance and data access.
e. What standards are you aware of that are not currently in wide use, but could improve data access and integration?	UDAP Dynamic Client Registration can greatly improve data access and integration. (https://www.hl7.org/fhir/us/udap-security/registration.html) (https://www.udap.org/udap-dynamic-client-registration.html) This terrific standard allows two systems to establish a secure connection directly without human holding up the process. Integration time can be reduced from weeks to minutes. This would allow a patient to pick a provider from all providers listed on their personal health app. Then the app would contact the provider's EHR directly and register as a client app by presenting the apps verifiable authentication certificate (x509 certificate) that shows they are part of an established trust network. The EHR can respond by issuing credentials to the app with access authorization. The app can now pass the patient's request for their records along with the patient's digital credentials. In addition to improving access, UDAP Dynamic Client Registration can greatly reduce costs for the provider and the app companies for manually managing these connections. It will also reduce or eliminate information blocking.

f. Are there redundant standards, protocols, or channels that should be consolidated?	
g. Are there adequate alternatives outside of TEFCA for achieving widespread patient access to their health information?	UDAP Dynamic Client Registration enables direct and secure, peer-to-peer, exchange of data. It is a great and inexpensive alternative to TEFCA. For this alternative to be successful, a provider endpoint directory and provider adoption would be needed.
PC–11. How are health information exchanges (HIEs) currently helping to advance patient access to health information in the real world?	HIEs are currently helping to advance patient access by allowing patients to aggregate their data from multiple healthcare systems, which lets patients see and manage their medical history in one place.
a. How valuable, available, and accurate do you find the data they share to be?	When the data that HIEs share is complete and current, it is extremely valuable to give patients and caregivers a more complete view of medical history. The availability of the data varies across HIEs, but in general, requires logins to different platforms and sometimes provider involvement, which can be barriers to access. Data from HIEs is generally accurate, but can be fragmented, outdated, or repeated.
b. What changes would you suggest?	We would suggest establishing a data standard for completeness, consistency, and usability across platforms and systems and also funding tools and integrations to help patient apps connect to HIEs more efficiently.
c. Are there particular examples of high-performing HIE models that you believe should be propagated across markets?	CRISP has a strong infrastructure and secure patient data access. It also supports social care integration through its community resource referral platform.
d. What is the ongoing role of HIEs amidst other entities facilitating data exchange and broader frameworks for data exchange (for example, vendor health information networks, TEFCA, private exchange networks, etc.)?	HIEs play a vital role by filling data gaps not covered by national or vendor networks, by enabling patient and caregiver access to usable health data, and by acting as intermediaries between local providers and national frameworks like TEFCA.

PC–12. What are the most valuable operational health data use cases for patients and caregivers that, if addressed, would create more efficient care navigation or eliminate barriers to competition among providers or both?	The most valuable operational health data use case for patients and their caregivers is being able to share relevant data (complete medical information) with providers and other care team members immediately when needed. This would greatly aid in enabling timely and accurate diagnoses and treatment. By extension, this would also reduce cost and health system burden by reducing the effort (manhours) it takes for a provider to collect health data and by reducing the risk of worsening medical conditions because of misdiagnoses and delay of care. True data portability, where patients and their caregivers hold all copies of their records and health information
3. Information Blocking and Digital Identity	
PC–13. How can CMS encourage patients and caregivers to submit information blocking complaints to ASTP/ONC's Information Blocking Portal? What would be the impact? Would increasing reporting of complaints advance or negatively impact data exchange?	CMS could make it more clear what information blocking complaints that CMS would enforce. We submitted a complaint on behalf of one of our members in 2022. The OCR response was unclear and left the patient feeling that the process was ineffective. CMS could also publicly list all organizations that have had information blocking complaints lodged against them and the status of those complaints. Even without aggressive enforcement, publicly listing complaint outcomes could improve compliance and restore patient trust.
PC–14. Regarding digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800–63–3 IAL2/AAL2 credentialing service providers (CSP)):	
a. What are the challenges today in getting patients/caregivers to sign up and use digital identity credentials?	This has not been a challenge for us. Patients/caregivers are willing to sign-up and use their IAL2 credentials when they have an obvious use for them. On OtisHealth, our members have been willing to sign up for their ID.me credentials when they learned they may use it to retrieve and consolidate their health records from different health systems.
b. What could be the benefits to patients/caregivers if digital identity credentials were more widely used?	Patients/caregivers would only have to remember one login credential (username and password). They would only have to update their address and contact information on one service. When they register with a new provider (or any service), the registration process can be simplified as ID.me can pass many registration details to the new provider (or company). Although here we are focused on the healthcare ecosystem, federated IAL2 credentials can be used in banking and many other industries.

c. What are the potential downsides?	Potential downsides include privacy concerns or hesitancy about third-party credentialing, though these can be mitigated through education and strong data protection standards.
d. How would encouraging the use of CSPs improve access to health information?	Increasing the use of CSPs can remove the “I forgot my password” issue and substantially increase the frequency that a patient utilizes digital health products to access their health data.
e. What role should CMS/payers, providers, and app developers have in adoption?	All members of the digital health ecosystem should be adopting IAL2 federated digital identity credentials. The more of us that participate, the more adoption will grow.
f. How can CMS encourage patients to get digital identity credentials?	Many patients already have federated digital IAL2 credentials such as ID.me. The providers accepting digital credentials would increase adoption. So, the best way forward would be to require providers to accept federated digital IAL2 credentials.

C. PROVIDERS	
4. Information Blocking	
PR–12. Should ASTP/ONC consider removing or revising any of the information blocking exceptions or conditions within the exceptions (45 CFR part 171, subparts B through D) to further the access, exchange, and use of electronic health information (EHI) and to promote market competition?	The ASTP/ONC should remove or revise information blocking exceptions. Specifically, a data resource holder (e.g. healthcare provider) should not block a patient’s access to their health records when the patient has presented their federated digital credential either directly from their personal health app or through a regulated network such as TEFCA. A data resource holder cannot use a selective rationale for not providing patient access when they are providing access to non-patients with lesser security measures.

E. TECHNOLOGY VENDORS, DATA PROVIDERS, AND NETWORKS	
1. Ecosystem	
TD–1. What short term (in the next 2 years) and longer-term steps can CMS take to stimulate developer interest in building digital health products for Medicare beneficiaries and caregivers?	CMS can stimulate developer interest by identifying population health targets and cost reduction opportunities specific to Medicare beneficiaries. Publishing regional, community-level, or provider-specific performance gaps would highlight areas ripe for digital health innovation. In the short term, CMS could offer grants to both providers and developers to accelerate deployment of solutions in those areas. Longer-term, CMS should ensure reimbursement structures consistently align with digital health workflows and use cases to support sustainable innovation.
TD–2. Regarding CMS Data, to stimulate developer interest—	
a. What additional data would be most valuable if made available through CMS APIs?	Patient-level data across encounters, diagnoses, labs, and medications—consolidated across sources—would be most valuable through CMS APIs.
b. What data sources are most valuable alongside the data available through the Blue Button 2.0 API?	Medication adherence and vitals data offer valuable insight into patient outcomes, such as avoidable readmissions and gaps in chronic disease management.
c. What obstacles prevent accessing these data sources today?	Currently, our only obstacle is funding the effort. A small grant and support for the first connection would be helpful.
d. What other APIs should CMS and ASTP/ONC consider including in program policies to unleash innovation and support patients and providers?	CMS and ONC should support APIs that enable developers to offer direct access to claims, clinical data, and care quality metrics—especially where provider portals or EHRs lack patient-facing capabilities. Short-term funding or voucher programs could help smaller teams build and maintain these APIs.

2. Digital Identity	
TD–3. Regarding digital identity implementation:	
a. What are the challenges and benefits?	Federated digital credentials (e.g., ID.me, Login.gov) streamline onboarding and improve patient matching by serving as a universal sign-on and update point for demographic data. A potential challenge is ensuring equitable access for populations without government-issued IDs or strong digital literacy. Standards and outreach must account for digital divide factors.
b. How would requiring digital identity credentials (for example, CLEAR, Login.gov, ID.me, other NIST 800–63–3 IAL2/AAL2 CSPs) impact cybersecurity and data exchange?	Requiring IAL2 digital credentials improves cybersecurity and trust in data exchange by ensuring each access request is tied to a verifiable identity. However, CMS should also monitor potential onboarding friction or disparities, especially in rural or aging populations.
3. Technical Standards and Certification	
TD–5. How could a nationwide provider directory of FHIR endpoints improve access to health information for patients, providers, and payers? Who should publish such a directory, and should users bear a cost?	A centralized directory would reduce fragmentation and boost trust by ensuring authoritative, up-to-date endpoint listings. CMS or ONC, with their oversight role, are best positioned to publish and maintain such a directory, and access should be free to encourage adoption.
TD–6. What unique interoperability functions does TEFCA perform?	TEFCA provides standardized data sharing rules to reduce fragmentation across EHRs, payers, and HIEs. TEFCA also provides a framework to cross-network exchange via QHINs, enabling broad access to health data.
a. What existing alternatives should be considered?	FHIR-based API direct connections via UDAP Dynamic Client Registration, HIEs, and private payer-provider networks.

b. Are there redundant standards, protocols or channels or both that should be consolidated?	TEFCA, EHR vendors, and payers often maintain separate endpoint directories. A single, CMS-supported FHIR endpoint directory would reduce fragmentation and support consistent access.
TD–7. To what degree has USCDI improved interoperability and exchange and what are its limitations?	Using USCDI has helped us be able to both establish a proper data structure and more quickly find mapping errors when ingesting data from other healthcare data sources.
a. Does it contain the full extent of data elements you need?	Yes.
d. Given improvements in language models, would you prefer a non-proprietary but less structured format that might improve data coverage even if it requires more processing by the receiver?	No. We prefer a structured format. We don't believe reducing structure requirements would correlate to improved data coverage. USCDI's structured format enables easier data mapping and reduced interpretation errors. While LLMs can augment unstructured content like clinical notes, we strongly prefer to retain structured standards like FHIR and USCDI to ensure data quality, integrity, and implementation simplicity.
4. Data Exchange	
TD–14. Regarding networks' use of FHIR APIs:	
a. How many endpoints is your network connected to for patient data sharing? What types, categories, geographies of endpoints do you cover? Are they searchable by National Provider Identifier (NPI) or organizational ID?	Our network connects to over 1,000 Epic endpoints using OAuth, HIEs via OpenID Connect (which connect to thousands of healthcare provider systems), and TEFCA via IAL2-enabled queries. Users search by provider name or region, depending on the data source. Patient records are returned without relying on NPI or org ID.
b. How are these connections established (for example, FHIR (g)(10) endpoints, TEFCA/Integrating the Health Enterprise (IHE) XCA, or proprietary APIs)?	We use FHIR APIs to establish connections with Epic endpoints. We use the standard TEFCA API integration and receive FHIR bundles though our QHIN, Health Gorilla.

c. Do you interconnect with other networks? Under what frameworks (for example, TEFCA, private agreements)?	We are on TEFCA as an Individual Access Services (IAS) vendor with Health Gorilla. We also have an agreement with Selfii to use their HIE access points through the Cures Gateway.
TD–16. What are the tradeoffs of maintaining point-to-point models vs. shared network infrastructure?	Point-to-point models can work if endpoints support dynamic registration and return clean data. Otherwise, they're costly and unscalable. Shared infrastructures improve reliability but may require policy shifts to ensure all providers participate and contribute meaningfully.
a. Do current rules encourage scalable network participation?	No. It seems many providers can opt out. They can also be non-compliant on access, data format, and responsiveness. More broadly, there may not be a business case (ROI) for a provider to participate in a network.
b. What changes would improve alignment (for example, API unification, reciprocal access)?	Reciprocal access could encourage more participation in networks.
TD–17. Given operational costs, what role should CMS or ASTP/ONC or both have in ensuring viability of healthcare data sharing networks, including enough supply and demand, that results in usage and outcomes?	CMS could support smaller providers' participation via reimbursement tied to data exchange metrics—such as query volume or successful responses—and consider one-time startup grants to address initial onboarding and credentialing costs.
5. Compliance	
TD–18. Information blocking:	
a. Could you, as a technology vendor, provide examples for the types of practices you have experienced that may constitute information blocking. Please include both situations of non- responsiveness as well as situations that may cause a failure or unusable response?	We asked two different health systems to accept our Epic production credentials to allow our patients, who are ID-verified with accredited IAL2 credentials, to retrieve their health records without having to log into the patient portal. After months of discussion and after completing a thorough security review, the health systems chose not allow access, despite acknowledging there was no additional security risk. In another case, a health system simply responded that patients may log in using portal credentials. And nothing else was required of them.
b. What additional policies could ASTP/ONC and CMS implement to further discourage healthcare providers from engaging in information blocking practices?	CMS could publish a list of providers with unresolved information blocking complaints. Public accountability, even absent formal penalties, may incentivize compliance and build patient trust in access rights enforcement.

END OF RESPONSE